Section 2.7 — Function Algebra — Answer Key

Math 103: College Algebra

1 Combined functions

Define the following functions:

$$\begin{aligned} f &= \{(-7,-9),(-1,-3),(1,\ \ 3),(5,\ \ 0),(9,-8)\} \\ g &= \{(-7,-4),(\ \ 1,-8),(5,\ \ 1),(7,\ \ 0),(9,-7)\} \\ h &= \{(-7,\ \ 0),(-1,-5),(1,-7),(7,-1),(9,-5)\} \end{aligned}$$

Find each of the following. If the given value is not in the domain of the given function, write "undefined."

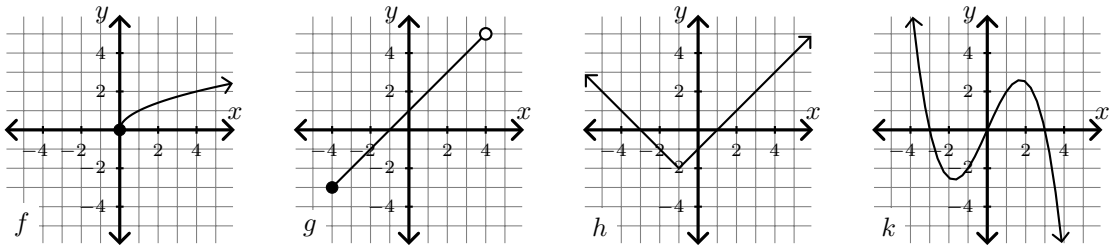
1. 56
2. undefined
3. -1
4. 0
5. -13
6. -11
7. undefined
8. 0
9. undefined
10. -1
11. $\frac{5}{3}$
12. -8

Find each domain.

13. $\operatorname{Dom} f = \{-7,-1,1,5,9\}$
14. $\operatorname{Dom} g = \{-7,1,5,7,9\}$
15. $\operatorname{Dom} h = \{-7,-1,1,7,9\}$
16. $\operatorname{Dom}(g+h) = \{-7,1,7,9\}$
17. $\operatorname{Dom}(h-f) = \{-7,-1,1,9\}$
18. $\operatorname{Dom}(f \cdot g) = \{-7,1,5,9\}$
19. $\operatorname{Dom}\left(\dfrac{h}{g}\right) = \{-7,1,9\}$

2 Combined Functions from Graphs

Consider the functions graphed below.



Find each of the following. If the given value is not in the domain of the given function, write "undefined."

20. -3
21. 2
22. 0
23. 0
24. undefined
25. undefined
26. 0
27. undefined

Find each domain.

28. $\operatorname{Dom} f = [1,\infty)$
29. $\operatorname{Dom} g = [-4,4)$
30. $\operatorname{Dom} h = \mathbb{R} = (-\infty,\infty)$
31. $\operatorname{Dom} k = \mathbb{R} = (-\infty,\infty)$
32. $\operatorname{Dom}(f-g) = [0,4)$
33. $\operatorname{Dom}(k+f) = [0,\infty)$
34. $\operatorname{Dom}(h \cdot k) = \mathbb{R} = (-\infty,\infty)$
35. $\operatorname{Dom}\left(\dfrac{g}{k}\right) = [-4,-3) \cup (-3,0) \cup (0,3) \cup (3,4)$

3 Combined Functions

$f(x) = 3x^2 - 8x + 4 \qquad g(x) = x^3 + 7x^2 - 4x - 28 \qquad h(x) = x - 2 \qquad k(x) = 3x^2 - 19x - 14$

Use the functions above to find the following.

36. -40
37. 0
38. 0
39. undefined

Write expressions for each of the following. Simplify your answer as much as possible.

40. $(f - k)(x) = 11x + 18$
41. $(g + f)(x) = x^3 + 10x^2 - 12x - 24$
42. $(f \cdot h)(x) = 3x^3 - 14x^2 + 20x - 8$
43. $(h - g)(x) = x^3 + 7x^2 - 5x - 26$
44. $(h + k)(x) = 3x^2 - 18x - 12$
45. $(f \cdot k)(x) = 9x^4 - 81x^3 + 122x^2 + 36x - 56$

Write expressions for each of the following. Do not simplify your answer.

46. $\left(\frac{f}{h}\right)(x) = \frac{3x^2 - 8x + 4}{x - 2}$
47. $\left(\frac{g}{h}\right)(x) = \frac{x^3 + 7x^2 - 4x - 28}{x - 2}$

Find each domain.

48. $\operatorname{Dom} f = \mathbb{R} = (-\infty, \infty)$
49. $\operatorname{Dom} g = \mathbb{R} = (-\infty, \infty)$
50. $\operatorname{Dom} h = \mathbb{R} = (-\infty, \infty)$
51. $\operatorname{Dom} k = \mathbb{R} = (-\infty, \infty)$
52. $\operatorname{Dom}(f + g) = \mathbb{R} = (-\infty, \infty)$
53. $\operatorname{Dom}(h - k) = \mathbb{R} = (-\infty, \infty)$
54. $\operatorname{Dom}(f \cdot h) = \mathbb{R} = (-\infty, \infty)$
55. $\operatorname{Dom}\left(\frac{g}{k}\right) = \left\{x \mid x \neq -\frac{2}{3}, 7\right\} = \left(-\infty, -\frac{2}{3}\right) \cup \left(-\frac{2}{3}, 7\right) \cup (7, \infty)$